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Martelage — the Evidence Package standard

Draft 0.2, document identifier `martelage 0.2`, wire format unchanged at `"v":1`. A directory format for records, plus one manifest that states the package's scope, its place in a package chain, and whether anyone other than the producer holds its digest.

- How to use it — verifying and producing a package (usage.html)
- The specification — `spec/MARTELAGE-0.1.md`
- The repository README — `README.md`

What an evidence package is

An evidence package is a **directory**, not an archive (§4.1 MA-16). It holds:

- **Record stream files.** JSON Lines files whose records are hash-chained within a stream: each record carries `seq`, which runs 1..n with no gaps, and `prev`, the `body_sha` of the record before it (§6, §7.1–7.2). Two writers' records are never merged into one stream (§6.6).
- **One manifest**, `package.json`, listing every other file in the package with its `sha256` and its byte length, and stating three things beyond that file list:
  - the package's **scope** — what it contains and, separately, what it does *not*, each absence carrying a meaning word from a fixed vocabulary (§8, MA-54 to MA-60);
  - its **place in a package chain** — `prev.package_sha` and `prev.how`, saying whether a predecessor exists and where that value came from (§7.4);
  - its **anchor statement** — whether a party other than the producer holds this package's digest, and under which named profile (§9). The slot has named profiles at this version, and one of them is an eIDAS qualified seal — the anchor profiles (usage.html#anchors) (§9.3).

The **package digest** is `sha256` over the exact bytes of the manifest. The manifest carries no digest of itself; because it lists every other file's digest, that one value transitively pins every file in the package, including the scope statement and the anchor claim (§7.3.1 MA-47).

All of it is checkable offline, with a POSIX shell and `sha256sum`, by someone who has never seen the producer: the zero-tooling check on the usage page (usage.html#zero-tooling) is a normative property of the format, not an example (§4.6 MA-21).

Why the standard exists: the honest sentence

The threat model comes first, because every mechanism in the standard is derived from it (§3). **The adversary is the operator.** The party that produced the package is assumed to have administrative control of the machine it was produced on, and can therefore rewrite the system's records, its auxiliary state, the

exported files, the manifest and the producing binaries — recomputing every digest inside the package correctly and consistently (§3.1 MA-10). Nothing inside a package proves anything against that party acting coherently; the standard calls this not a defect to be designed away but the residual every mechanism is measured against (§3.2 MA-11).

So the standard states its own residual, and every conforming validator MUST print the following two sentences verbatim on every run *that names a package* — including a run that reports the package conformant, and including a run that ends at exit code 2 because the package could not be read (§3.3 MA-12):

> A package proves that what it contains was not altered after it left; it cannot prove that what it contains is all there was, and against an operator who rewrote everything before the first anchor it proves nothing.
> A coherent tamper performed before the first observation is invisible to every mechanism in this standard.

That is the whole position. A format that names its adversary can be argued with; a format that does not cannot (§3).

MEASURED on this tree: the reference validator prints both sentences, word for word, in a `residual` block on every run that names a package — the usage page pastes the block from a real run (usage.html#verify).

The five mechanisms, and exactly what each buys

Each mechanism is specified and reported separately, because their powers differ by orders of magnitude and a reader will otherwise credit the weakest with the strength of the strongest (§3.4). This is the standard's own table:

The five mechanisms, each with its exact power (§3.4)

#	Mechanism	What it buys	What it does not
i	Internal consistency (§5.1–5.2, §6.2, §7.1–7.2)	A partial or sloppy rewrite shows as a break: a moved body digest at a fixed position, a gap, a count that disagrees with the payload.	A wholesale, internally consistent rewrite passes.
ii	Staleness against the previous package (§7.4)	Converts an invisible omission into a break visible one package later — if someone kept the earlier package.	An operator who starts a fresh chain and was watched by nobody.
iii	Witnesses outside the audited subject (§3.6)	Forces the operator to rewrite *more* places consistently.	Stops the operator. Each extra rewrite is another place to contradict a copy someone else holds — it multiplies cost, never prevention.
iv	The enrollment gap (§3.5)	Nothing. It is named so it is not hoped away.	—
v	The external anchor (§9)	Proves a digest existed at a time, held by someone other than the operator.	Truth, completeness, or that the operator told the witness everything.

The standard also gives the one-line reading of its own layers, which is the sentence to carry away (§7.6 MA-53):

> Layers 0–2 stop tampering *with a package you hold*; layer 3 bounds tampering *across packages a verifier held*; only layer 4 binds the package to a party outside the operator's reach.

Two consequences are stated as requirements rather than left to inference. A validator report MUST show which layers were armed, and MUST NOT let an unarmed layer read as a pass (§7, §10.4 item 6). And a chain proves order and integrity over what is *present*: it does not prove completeness — a consistently-chained prefix, and a consistently-chained subset, pass every check in layers 0 and 1 (§7.2.3 MA-45).

What it is not

- **Not a record format.** It does not say what an auditable system should record, in what schema, or with what semantics. It defines the outer layer that carries someone else's records; the record grammar is the payload's own business, named and not specified here (§0.5 MA-5).
- **Not a claim about the records.** A conformant package may contain records that are false, and may omit records that existed. Conformance says the package is well-formed and self-consistent; it says nothing about truth or completeness (§2.2 MA-7).
- **Not a certification, and not a registry.** No registration, no permission and no fee is required. A third party claims conformance by publishing a statement in its own name, naming the requirements it relies on and any it does not satisfy, with the reason; the standard certifies no one (§2.4 MA-9).
- **Not a service.** The reference validator runs offline with a POSIX shell and the utilities §10.1 names: no network, no interpreter beyond those, no key material, no service, no producer (§10.1 MA-68).
- **No keys, and no third party required for the base tier.** A package with no third party, no seal, no timestamp and no key material is fully conformant; nothing in the conformance definition requires any of them (§9.1(a) and (b), MA-61). The anchor is an optional slot, not a tier.
- **Not a ladder.** There is one level, deliberately: a ladder reads as "the lower tier is deficient", which is backwards, since the unanchored tier is the adoption surface (§2.3 MA-8).
- **Not a promise about the producer.** A validator is not required to know a producer; `producer` is a claim under the producer's own name (§10.8 MA-75).

One implementation is named in the standard's Annex A, and only there. That annex is informative: nothing in it is required to conform, and removing it would leave the standard whole.

The name

Martelage is the forestry marking of trees with a hammer, by a party other than the owner: a durable mark, attributable to whoever struck it, that decides which trees are retained. The standard's own text does not gloss its name anywhere (MEASURED: no occurrence of "forestry" in `spec/MARTELAGE-0.1.md` or `README.md`); the sentence above is the name's plain sense, an INTERPRETATION, and no clause of the standard depends on it.

Status

- **Draft 0.2** — published for adoption and criticism; not yet a stable standard. A change to this document's requirements may be breaking while it is a draft; a change to the published wire format (`"standard":"martelage","v":1`) must not be (§11.6 MA-81). The bump from the first draft to this one is recorded in §11.7 (MA-83): the verifier surface grew and the wire format did not, so a package written for the first draft is a `0.2` package.
- **Every normative requirement carries an identifier**, `MA-1` … `MA-83`, so a conformance report can cite what it reports as met or unmet; a clause that cannot be cited cannot be tested (§0.3 MA-3). MEASURED: all 83 identifiers appear in the specification.
- **Licence split:** the specification text is CC BY 4.0; the reference validator and all code in the repository are Apache-2.0 — see `LICENSE`.
- **Where it lives:** the specification is `spec/MARTELAGE-0.1.md`; the reference validator is `validator/martelage-verify` and `validator/p3-verify`; the fifty-eight test vectors are under `vectors/`, each with an `expected.txt`.

Martelage — using the standard

Draft 0.2, document identifier `martelage 0.2`, wire format unchanged at `"v":1`. Verifying a package you were handed; producing one from your own records; and what a validator must admit it cannot check.

- What an evidence package is, and what it does not prove (index.html)
- The specification — `spec/MARTELAGE-0.1.md`
- The repository README — `README.md`

Every command on this page was run by us on this tree (HEAD `90dbceb`) and its output pasted verbatim. Blocks marked MEASURED are copied from those runs; a statement that is a reading of the specification rather than a run is marked INTERPRETATION, in the manner the standard itself requires (§0.4 MA-4).

Verify: the common case

You have a package directory. You run the reference validator against it. That is the whole procedure — there is nothing to install, no account, no key, no network:

```
sh validator/martelage-verify vectors/01-minimal-conformant/pkg
```

MEASURED, exit code `0`. This is the run's standard output in full, unedited; nothing was written to standard error. Its first line names the document version the base validator was written against — `martelage-verify` is unchanged by the bump to `0.2`, because the wire format did not move (§11.7), and the banner is quoted here as the tool prints it rather than edited to match the page:

```
martelage-verify — MARTELAGE-0.1 reference validator

package      vectors/01-minimal-conformant/pkg
digest       ab2bc3a563e26e1454400ba70cebad73a849d194bcd2c1d2787c0efaf3c2653c   (sha256 of package.json: this is the
↳ package digest)
standard     martelage v1 (kind package-manifest)
subject      c419edf957fda5f89c2988273d3d00e9e5b2a69ebe7f6bca22644da52bc129f3 (kind note-log)
producer     example / by-hand 0 / contract martelage-example-notes 1
created_at   1700000200
prev         null (how: none)
files        2
records      2 over 1 stream(s)

scope
  contains:
    - 2 record(s) on stream(s): note
  not_contains:
    - anything not listed above [not-recorded] this package is a hand-made example for the standard

files
  OK   records.jsonl                abf920fbb49003e1b06badc085e74f8396713bca64d2f123eccdb256e6a36bad  776 bytes
↳ role=records
  OK   notes.md                     6ed0b5af646aad99f7ae97d94c2a56623b36de11d5967af4fc508bececdbca10  55 bytes
↳ role=document

streams
  FILE                STREAM           N     HEAD           VERDICT
  records.jsonl       note             2     6e28c56d1a81   OK
```

```
anchor
  NOT ANCHORED — this package's digest is held by no one but its producer

layers
  layer 0  body digests        checked
  layer 1  stream continuity   checked
  layer 2  package digest     ab2bc3a563e26e1454400ba70cebad73a849d194bcd2c1d2787c0efaf3c2653c
  layer 3  package chain       not armed (no --prev-package given) — omission across packages is NOT checked
  layer 4  anchor              not claimed by this package

refusals
  none

findings
  none

ignored
  2 field(s) this validator does not know, counted per (object, field) — ignored, never guessed at (§5.2.11 MA-33,
↳ §11.5 MA-80):
  'records.jsonl' line 1     generator
  'records.jsonl' line 1     counts

not decidable
  nothing — every check this validator implements could reach a verdict on this package (MA-70)

residual
  A package proves that what it contains was not altered after it left; it cannot prove that what it contains is all
↳ there was, and against an operator who rewrote everything before the first anchor it proves nothing.
  A coherent tamper performed before the first observation is invisible to every mechanism in this standard.

verdict      CONFORMANT
```

Read the block in this order and the report tells you what it did and did not do. The `layers` block names each layer and whether this run armed it. The `anchor` block says, in words, that no one but the producer holds this digest. The `residual` block is the §3.3 sentence, printed on every run that names a package. The `ignored` and `not decidable` blocks are sections of their own, so that a field the validator did not understand and a check it could not reach are never invisible (§10.4 items 9 and 10, MA-71).

A refused package reads the same way, with the failing check named — MEASURED on `vectors/03-stream-gap/pkg`, an interior record removed, exit code 1. Trimmed here to the refusal and the verdict; the rest of that report has the shape shown above:

```
refusals
  payload: records.jsonl stream 'note': seq 4 where 3 was expected;

verdict      NON-CONFORMANT
```

Exit codes

The exit code is the verdict; the tail of the output is not. The four codes are the standard's (§10.3 MA-70, and `README.md`):

The exit codes (§10.3)

code	meaning
`0`	**CONFORMANT** — every check the run could perform passed. An unanchored package exits 0; the NOT-ANCHORED line has been printed.
`1`	**NON-CONFORMANT** — a check failed: a digest, a length, a gap, a broken predecessor link, a torn line, a manifest count, an unknown scope word, an unsupported version, a chain break against `--prev-package`, a claimed anchor that is malformed or that the evidence contradicts.

code	meaning
2	**USAGE OR UNREADABLE** — the invocation is wrong, or an input could not be read (an unreadable file is not an absence).
3	**CANNOT DECIDE** — an input the validator needs to decide is absent: anchor evidence for a claimed anchor, trust anchors for P3, or a profile-specific verifier this validator is not. The validator MUST name the missing input.

Why code 3 exists. An instrument that cannot return "undecided" returns something else instead. A claimed anchor that cannot be checked is neither conformant nor refuted, and reporting it as either would be a false statement; it is not conformant enough to pass, so 3 is non-zero (§10.3). This is not a nicety — it is the same discipline as the residual: the format says what it could not establish.

Verify without any tooling

The manifest is canonical and lists every file with its digest, so a verifier with nothing but a POSIX shell can check the file set without any validator. This one-liner is a normative property of the format: a manifest whose canonical form does not admit it is non-conformant (§4.6 MA-21).

```
grep -o '{"path":"[^"]*","sha256":"[0-9a-f]*"' package.json |
  sed 's/{"path":"\([^"]*\)","sha256":"\([0-9a-f]*\)"/\2  \1/' > SHA256SUMS
sha256sum -c SHA256SUMS
```

MEASURED, run from inside a copy of `vectors/01-minimal-conformant/pkg` made in a temporary directory — the one-liner writes `SHA256SUMS` into the current directory, so do not run it inside the repository unless you mean to add that file. The three lines below are the written `SHA256SUMS` followed by `sha256sum -c`, exit code `0`:

```
abf920fbb49003e1b06badc085e74f8396713bca64d2f123eccdb256e6a36bad  records.jsonl
6ed0b5af646aad99f7ae97d94c2a56623b36de11d5967af4fc508bececdbca10  notes.md
records.jsonl: OK
notes.md: OK
```

What that check covers is the file set: every listed file present, and each file's bytes equal to the digest the manifest states. What it does not cover is everything above the file list — the record envelopes, the per-stream sequence and predecessor chain, the manifest counts, the scope vocabulary and the anchor claim (INTERPRETATION, from §4.4 and §7). For those you need the validator.

Produce: what a second project must build

There is no library and no service. Production is decentralised and cheap; checking is centralised and shared — an adopter does not reimplement the validator, it runs the published one, or reads it and writes its own against the specification (Annex F). A

project that already writes dated records to durable files needs exactly two things:

1. **An emitter** that puts its records into the envelope of §6.2: canonicalise each body, compute `body_sha`, assign `stream` and `seq`, wire `prev`, print the stream manifest first. For a project whose records are already one JSON object per line, Annex F prices this at roughly 80–120 lines in any language.
2. **A packager** that lists its files with digests, writes a canonical manifest with an honest scope, a `prev` from a one-line state file, and `anchored:false` — Annex F prices it at roughly 60–100 lines.

Those two line counts are the standard's own estimate (Annex F), not a measurement of any particular project; treat them as a PROJECTION in the §0.4 sense.

Annex E is a worked package written out by hand with `printf` and `sha256sum`: two files, one shell, no network, no keys, no service. It is the whole of what a second project must produce to start, and a stranger can copy it. `README.md` records that executing that example verbatim, in an empty directory, with nothing from the repository on the path except the validator, produces a package the validator reports CONFORMANT (exit 0) — that is the README's measurement; this page did not re-run it.

Do not expect a validator to live inside a package. A package is not required to carry one, and a validator found inside a package proves nothing about that package, because it sits inside the directory the operator can rewrite; checkability comes from the published validator or from the verifier's own implementation, obtained through a channel the operator does not control (§4.5 MA-20).

Chain and anchor: the two layers that look outside the package

Two verifier inputs arm the two layers that reach beyond the package directory (§10.2 MA-69). Neither is required for the base tier — and an unarmed layer must never read as a pass (§7, §10.4 item 6).

`--prev-package DIR` — arms layer 3

Without it, a record dropped between two packages is invisible: the omission happened outside the package you hold, and nothing inside this package records that it ever existed. MEASURED, both arms on `vectors/11-chain-omission` — first the successor alone, exit code `0`, with the layer reported as unarmed. Trimmed to the `layers` block and the verdict:

```
layers
  layer 0  body digests        checked
  layer 1  stream continuity   checked
  layer 2  package digest     20be1945067e73fbffebabd55d2408d1241c74db7a934304bd49b83d6c72ecf7
  layer 3  package chain       not armed (no --prev-package given) — omission across packages is NOT checked
  layer 4  anchor              not claimed by this package
```

```
verdict      CONFORMANT
```

Now the same package with its predecessor supplied — the same omission, exit code 1. Trimmed to the refusal and the verdict:

```
sh validator/martelage-verify --prev-package vectors/11-chain-omission/pkg-1 vectors/11-chain-omission/pkg-2
```

```
refusals
  chain: stream 'note' is not a prefix of the same stream in the supplied previous package — records were dropped,
↳ re-ordered or rewritten across packages (§7.4.2 MA-50)

verdict      NON-CONFORMANT
```

That is the whole power of the layer, and it is bounded exactly as §7.4.3 says: it converts a silent omission into a break visible one package later — *if someone kept the earlier package* — and it does not stop an operator who starts a fresh chain and is watched by nobody (§7.4.3 MA-51).

`--anchor-evidence HEX|FILE` — arms the anchor

"Anchored" is a property of (package, verifier's evidence), not of the package alone: the same package is verified-anchored for a verifier who brought the holder's digest, and cannot be decided for one who did not (§9.5 MA-66). MEASURED on `vectors/04-anchor-unverifiable`, a package claiming a p1 anchor with no evidence supplied — exit code `3`. Trimmed to the `anchor` block, the `not decidable` block and the verdict:

```
anchor
  CLAIMED — profile 'p1-second-party-digest', binds 'manifest-digest'
  claim: holder 'witness@example-archive', channel 'ask witness@example-archive for the digest of this package'

not decidable
  anchor: profile p1 is claimed but --anchor-evidence was not supplied — the holder's digest is the input this check
↳ needs (§10.3 MA-70)

verdict      CANNOT DECIDE — 1 input(s) this validator needs were absent
```

And the same claim with the holder's digest supplied (`vectors/05-anchored-verified/evidence.txt`) — exit code `0`. Trimmed to the `anchor` block and the verdict:

```
sh validator/martelage-verify --anchor-evidence vectors/05-anchored-verified/evidence.txt
↳ vectors/05-anchored-verified/pkg
```

```
anchor
  ANCHORED — profile 'p1-second-party-digest', binds 'manifest-digest'
  claim: holder 'witness@example-archive', channel 'ask witness@example-archive for the digest of this package'
  evidence: the holder's digest equals this package's digest — VERIFIED (§9.3 MA-64)

verdict      CONFORMANT
```

Which profile a package names decides what a verifier can do with its anchor, and one of the three is an EU qualified seal. The profiles have their own section.

The anchor slot: three named profiles

The anchor is a slot with named profiles, not one vendor: §9.3 names three at this version and leaves the slot open to others

(§9.1(c) MA-61). It gets a section of its own because layer 4 is the only layer whose witness the operator cannot recompute (§7.5 MA-52) — and because the profile a package names is the thing that decides what a verifier can actually check. What no profile changes is the claim's force: an anchor never says the record is true or complete, only that a digest existed, at a time, by someone else (§3.7 MA-15, §7.5 MA-52).

The three anchor profiles at this version (§9.3)

profile	wire string — the value of `anchor.profile`	what the base validator does with it
P1	`p1-second-party-digest`	Verifies it fully: the holder's digest must equal the package digest (§9.3 MA-64).
P2	`p2-append-only-log`	Compares a digest the log states to the package digest. A digest that contradicts the package is a refusal (exit 1); a digest that matches still leaves inclusion undecided, so it reports exit 3 and names what it could and could not check (§9.3, §10.3 MA-70).
P3	`p3-eidas-qualified-seal-asic-e`	Cannot verify a signature, so it reports exit 3 and names the profile-specific verifier it is not (§9.3, §10.3 MA-70).

`P1`, `P2` and `P3` are labels for those strings, never values: a manifest writing `P3`, or the profile's name in any other spelling, is refused as an unknown profile (§9.2 MA-63, §9.4 MA-65). The wire string is what a validator matches.

`p3-eidas-qualified-seal-asic-e` — the EU qualified seal

P3 is the profile for a deployment that already has to answer to EU law. In the standard's own terms it is an ETSI EN 319 162-1 ASiC-E container holding the package files with a CAdES or XAdES signature over their digests, created by a legal person *other than the operator*, with a qualified electronic time stamp and optional long-term validation (§9.3 MA-64). The party who seals is the point: a seal is only an anchor because someone the operator is not made it.

The claim is a reference, not the seal. A manifest cannot carry a value derived from its own digest (§9.1a MA-62), so the container and the trust anchors stay on the verifier's side and the claim only says where to get them. This is the claim shape, verbatim from §9.3:

```
{"container":…,"sealer":…,"seal":"qualified-electronic-seal",
 "timestamp":"qualified"|"none","ltv":<bool>}
```

What the reference validator does with that claim is nothing, and it says so. It cannot verify a signature — that needs a signature verifier and trust anchors — so it reports exit code `3` CANNOT DECIDE and names the profile-specific verifier requirement (§9.3, §10.3 MA-70). That is the standard's own rule applied rather than an omission: a claimed anchor the verifier cannot verify MUST NOT pass (§9.1(e) MA-61). Nor does a specialised verifier get a lighter contract for being specialised — a profile-specific verifier MUST report under §10, the same exit codes and

the same report items (§9.3, §10.4 MA-71) — so "we sealed it" is not a way to skip the report.

Status: built, by a second implementation. The base validator still stops at exit code `3` and still names the profile-specific verifier it is not — that has not changed and is not going to, because §9.1(b) keeps keys and trust anchors out of the base tier. The verifier it names is `validator/p3-verify`: a second implementation that shares no code with `martelage-verify`, runs under `python3` and calls the `openssl` binary for every cryptographic operation. It has no network path at all: it never downloads anything, never opens a URL, never writes an OCSP request and never contacts an OCSP responder. Revocation is checked only against the material the person running it hands it — `--revocation FILE` for a CRL, `--ocsp-response FILE` for an OCSP response, `--revocation-dir DIR` for a directory of both — and without any of them revocation is not checked and the report says so.

```
validator/p3-verify \
  --container vectors/34-seal-verified/seal.asice \
  --trust-anchors vectors/pki/anchors \
  --operator-cert vectors/pki/operator.crt.pem \
  vectors/34-seal-verified/pkg                    # exit 0
```

It takes the container and the anchors on the verifier's side, as §10.2 (MA-69) says a P3 input must be, plus inputs of its own: `--operator-cert`, the operator's certificate, and optionally revocation material — `--revocation`, `--ocsp-response` or `--revocation-dir`. Without the operator certificate the "legal person *other than the operator*" clause cannot be checked, so the run reports `3` for that clause and says the input is missing. It never assumes the signer is someone else — which is why no P3 run reaches exit `0` without it.

What `--revocation` takes, and what it establishes. A CRL, PEM or DER, read from a file the verifier was given; repeat it once per CA in the chain, because a CRL speaks only for the certificates its own issuer signed. With it the verifier checks its own signature against a certificate the run holds, checks that its window (`thisUpdate` … `nextUpdate`) contains now, and checks every certificate in the established chain that the CRL speaks for: a listed certificate **refuses** the container and the report names the certificate, the CRL, the revocation date and the reason. A certificate no supplied CRL speaks for is reported as *not checked* and makes the run `3` — never "not revoked". A CRL that has expired, is not yet valid, or whose signature does not verify is reported unusable and the run does not claim revocation was checked.

What `--ocsp-response` takes, and what it establishes. An OCSP response — DER, or PEM with the standard header — read from a file the verifier was given and read with `openssl ocsp -respin`. No request is written, no responder is contacted: the response is material you were handed, and this

tool has no network path at all. It checks the response's own signature against a signer the *certificate's issuer* authorised — the issuer itself, or a responder the issuer issued — and requires that signer's chain to reach a supplied anchor; and it requires the response's CertID to name the certificate being checked, because an OCSP answer is about one certificate and never about a set. A `revoked` status **refuses** the container (`1`), naming the certificate, the responder, `thisUpdate`, `nextUpdate`, the revocation time and the reason. A `good` status is **checked and clean for the certificate the response names**, and is never carried across to another one. `unknown`, an expired or not-yet-valid window, an unverifiable signer, an unsuccessful response status, or a response that names no certificate of this chain is its own stated outcome and makes the run `3` — none of them reads as a pass.

What `--revocation-dir` takes. A directory of revocation material, so a chain does not need one flag per file. Every regular file in it that parses as a CRL or as an OCSP response is applied by the rules above, and the report names, per certificate, which supplied file covered it. A file that parses as neither is **reported and makes the run `3`**: a verifier that silently skips material it was handed is claiming more than it checked. It is a separate flag from `--revocation` because that flag's published contract is a single file — a path that turns out to be a directory should be an error, not a different check. The three flags combine, and the coverage rule does not soften: every certificate in the chain must be spoken for by something, or the run is `3` naming what is uncovered. What remains uncheckable is said in the same breath: no OCSP response is ever *requested*, no CRL is ever downloaded, a revocation published after a CRL's or a response's `nextUpdate` is invisible, a supplied status is the status now and not at the claimed time, and long-term validation is still never validated.

Where the material may come from: the fetcher. Material does not have to be handed over by hand. `tools/p3-fetch` is a **fetcher** — a transport tool that sits *beside* the verifier and is not part of it (§9.3.2, MA-85). Given a certificate, it reads the OCSP URL from the certificate's AIA and the CRL URLs from its CRL distribution points (or takes explicit URLs), fetches the material over `http` or `https`, and writes it — exactly the bytes that answered, never re-encoded — into a directory `--revocation-dir` accepts unchanged, with a provenance record (every hop, the TLS peer's certificate hash, whether the nonce it sent came back, the sha256, the retrieval time) written *beside* the directory, never inside it. It refuses before writing anything that does not parse, that is a delta CRL, that answers a responder error (`tryLater`, `unauthorized`), that redirects to a scheme it does not fetch, that outruns its redirect cap, its byte cap or its timeout, or whose TLS peer does not verify under the anchor directory it was given. The transport sentence, stated plainly: **a fetch is transport, not trust** — it establishes where bytes came from and nothing about what they mean. It establishes no freshness, no honesty of the responder, no identity

between the URL's owner and the CA, and no status at any time outside the window the material itself states; every run of the tool says so. It is not a validator and reports no verdict — the verifier decides, by re-making every check it makes on material handed over by any other means, which its own report states (*"read from disk, offline: nothing is fetched"*). And one boundary does not move: fetched material can never satisfy an `ltv:true` claim (§9.3.1, MA-84), because that claim asserts evidence the container carries inside the seal, and no directory of bytes fetched at check time carries anything. Vector 66's pair arm runs the whole path — fetch over a loopback rig, then verify the fetched directory — and the run is CONFORMANT only because the verifier's own checks say so.

The clause is checked over the whole signer set. A signature entry may carry more than one signer, and the operator only has to be one of them. The verifier enumerates *every* signer of *every* entry from openssl's `-signer` output, holds that count against the signerInfos the entry itself states, refuses the container if the two disagree, and compares every one of them with `--operator-cert`: if any signer is the operator, the run refuses and names it. The report prints the signer count and every signer's subject, window and public-key digest — never a single subject standing in for "the signer". The same set is what the certificate-date check runs over: every signer, every certificate in the chain each signer reaches, and the timestamping authority's own chain, all held to the time the verified token establishes and never to this host's clock.

What `p3-verify` checks, and the four exit codes (§10.3 MA-70)

code	meaning for P3
`0`	**CONFORMANT** — the signature entry is the construction this verifier implements (a detached CAdES over the manifest), the container is well-formed (no duplicate entry name, every covered path inside the package), the CAdES signature verifies over the ASiC manifest, *every* signer of *every* signature entry chains to a supplied anchor and none of them is a CA, every sealed digest is the package's, the RFC 3161 token verified, every certificate the run established — every signer, the whole chain, the TSA's own chain — was valid at the claimed time, no signer's key is the operator's, and — *only if* `--revocation` *was given* — every certificate a supplied CRL speaks for is absent from it. Revocation unarmed is not part of this verdict and is reported as unarmed.
`1`	**NON-CONFORMANT** — a signature entry whose content is **embedded** rather than detached over the manifest: a shape this verifier does not implement, refused by name before anything else is accused, together with the EN 319 162-1 concatenated-digest form and XAdES; a malformed container (a wrong `mimetype`, a duplicate ZIP entry name, a covered path that leaves the package, an entry the manifest lists and the ZIP does not hold, a sealed digest that no longer matches the file on disk), a signature that does not verify, a signer certificate that is a CA, any signer or chain certificate that was not valid at the claimed time, a token from a timestamping authority that was not, a seal made with the operator's own key *or co-signed with it*, a certificate a supplied CRL lists — or a supplied OCSP response reports — as revoked, or `--require-timestamp` given with no token.
`2`	**USAGE OR UNREADABLE** — a bad invocation, an unreadable input, or a package that does not claim P3 at all (that package is the base validator's).
`3`	**CANNOT DECIDE** — an input was not supplied or cannot be used: the container, trust anchors, `--operator-cert`, a verified timestamp to establish the claimed time, or revocation material that has expired, is not yet valid, whose signature does not verify, that answers `unknown`, that parses as neither a CRL nor an

code	meaning for P3
	OCSP response, or that speaks for only part of the chain. A signature that verifies but reaches no supplied anchor is *this* code, not a refusal: the verifier was not given the anchor that decides it (§9.5 MA-66). So is a certificate no supplied material speaks for: "nothing said" is not "nothing revoked".

What it still cannot check, on every run. The verifier prints a section headed `not checkable by this verifier (every run)` before its verdict — and prints it to stderr on an exit-2 run as well, so "on every run" is true of the runs that have no report — and the list below is that section's content, and it is printed in the state the run is actually in: **revocation** — unarmed, the entry says no CRL and no OCSP response is read and no OCSP responder is contacted, so "valid at the claimed time" means the certificate dates and never revocation status; armed, it says exactly which material was used, with each CRL's issuer, `thisUpdate`, `nextUpdate` and entry count and each response's signer, status and window, that a certificate no supplied material speaks for has no revocation status here, that a revocation published after a CRL's or a response's `nextUpdate` is invisible, and that a supplied status is the status *now* and not at the claimed time; **long-term validation beyond what the container carries** — the material the container carries *is* validated (§9.3.1 MA-84, below), and what stays outside is a validation value the container does not carry (nothing is fetched), the preservation of the package by an archive authority over future years, whether the archival timestamps' own authorities are themselves revoked, and the revocation of a supplied trust anchor, which is not an omission to repair: to be revocable an anchor needs a root of trust above itself, and that regress is what the anchor exists to stop; the **truth and completeness** of the records, which is limit (iii) below and is kept distinct from the residual of §3.3; the **signer's legal identity** beyond the subject its certificate claims; whether the certificate was **issued for sealing** at all (this standard defines no key-usage profile for a seal, so the chain is verified with `-purpose any` and that is said rather than assumed — the `basicConstraints` CA bit is the one key-usage-shaped property that *is* checked, because a CA is not a sealing party); and a **re-forged history**, limit (ii). A run that could not check something never reads as having checked it, and when no verified time exists the date question is reported CANNOT DECIDE rather than skipped.

What it is not. Not a qualified trust service, not an eIDAS certification, and not a claim of compliance with ETSI EN 319 162-1. The container shape it verifies is the manifest-signature form — a detached CAdES over `META-INF/manifest.xml`, carrying the covered files' digests in that manifest — and the tool says so in its own report, because the EN 319 162-1 multi-file concatenated-digest form is not what it parses and it will not report that as verified. A container of another shape is **refused** rather than absorbed: an embedded-content CMS, the

concatenated-digest form and XAdES are all shapes this verifier does not implement, and a run that meets one says which shape it met instead of reporting it as a detached signature it is not. It does not verify XAdES. It is not a revocation service either: `--revocation`, `--ocsp-response` and `--revocation-dir` check the material they are handed and nothing else — no live OSCP, no CRL download and no OCSP request anywhere. Long-term validation is validated **for the material the container carries**, under the definition of §9.3.1 (MA-84) and its stated limits; a value the verifier was not given is not fetched, and every run says so. It does not make the base validator verify seals, and it does not change the base tier's tool list (§10.1 MA-68).

Long-term validation, validated (§9.3.1 MA-84). When the claim states `"ltv":true`, the container must *carry*, under `META-INF/ltv/` and covered by the ASiC manifest so the seal binds it, revocation evidence — a CRL or an OCSP response — for **every** certificate of the chain the run established, and for each value an RFC 3161 **archival timestamp whose signature covers that value's exact bytes**. The run checks, for each carried value, that its own **signature verifies** under the certificate that issued it; **which certificate of the chain it speaks for**; that it was **inside its window at the time its own archival timestamp establishes** — that time, never this host's clock; and that the **archival timestamp's signature verifies**, that its imprint covers the value's bytes, that its chain reaches a supplied **anchor**, and that its own authority was valid at the time it states. Four outcomes, and only one may pass: complete, valid and correctly timed material ⟶ the run may report conformance and **names every value it validated**; absent or incomplete material ⟶ refused, naming what is missing; material present but expired at the archival time, unverifiable, revoked, or speaking for nothing ⟶ refused, naming the value; and a time that **cannot be established** ⟶ CANNOT DECIDE, with the window check said out loud rather than skipped. `"ltv":false` reports that no LTV was claimed, and material carried under `"ltv":false` is a **finding** that leaves the exit code alone.

The fixtures are a **throwaway** test PKI under `vectors/pki/` — two roots, an intermediate, a second intermediate with the same key and subject that expired in 2021, a sealer, a second sealer under the same intermediate, a sealer under the withheld root, an expired sealer, the operator, a timestamping authority and a second TSA certificate with the same key and subject that expired in 2021 — with its private keys committed on purpose, because a vector that cannot be rebuilt from what is in the tree is not reproducible. The dates are fixed with `-not_before`/`-not_after`, so the expired-signer, expired-intermediate and expired-TSA vectors are expired on any host at any time; the keys are regenerated on each run, so `sh tools/mkpki.sh && sh tools/mkp3.sh` rebuilds the PKI and the containers together. The CRLs those vectors hand the

verifier are built by `tools/mkcrl.sh` — one per CA in the chain, plus a revoked entry, a window that ended in 2021 and a CRL from a CA that issued nothing in the chain — and the OCSP responses by `tools/mkocsp.sh`, which signs them with a responder intermediate A issued and, for the unanchored arm, with one that is self-signed. Those responses are then re-dated and re-signed by the same responder key so their windows are the same fixed dates the CRLs use: `openssl ocsp` stamps a response with the build second and refuses a non-positive window, so a shipped "expired response" is impossible without that step. So The archival timestamps the long-term-validation vectors carry are built by `tools/mklv.sh`, one RFC 3161 token over each CRL and over the OCSP response those vectors carry, with `openssl ts -query`/`-reply` and no responder anywhere; they are not re-dated, because a token's genTime lives inside a CMS SignedAttributes structure a script cannot move, so what each arm asserts is relative to the build second and the fixture dates fix the direction of the inequality.
`sh tools/mkpki.sh && sh tools/mkcrl.sh && sh tools/mkocsp.sh &&`
`sh tools/mklv.sh && sh tools/mkp3.sh`
rebuilds the PKI, the revocation material, the archival timestamps and the containers together. All of them run offline, open no port and need only `openssl` and `python3`. Thirty-two vectors (34–65) cover this profile, each with an arm that passes and an arm that fails, differing by one thing: a digest that moved, an anchor that is withheld, an operator that sealed, an operator that *co-signed*, a CA used as the sealing party, an intermediate or a TSA that expired, a ZIP entry name that appears twice, a covered path that leaves the package, a signature whose content is embedded instead of detached, a sealer on a CRL, a CRL that lists nobody, a CRL from another CA, a CRL that has expired, a sealer on an OCSP response, a response that says good, a response whose window ended in 2021, a response signed by a responder that chains to nothing, a directory of both kinds in which one certificate stays uncovered, and a directory holding one file that is not revocation material at all. Seven more (59–65) cover long-term validation: complete and correctly timed material, an empty container under `ltv:true`, a carried CRL that was out of its window at the archival time, carried material that leaves one certificate of the chain uncovered, the same material under `ltv:false` (a finding, exit unchanged), a carried archival timestamp whose own signature does not verify, and a carried CRL that lists the sealer as revoked.

§9.3 requires three limits to be stated wherever P3 is offered. They are the profile's terms, not its small print:

- **(i) A seal made with the operator's own key is worthless against the operator**, because they hold the key. That is why the verifier compares *every* signer of *every* signature entry: a co-signer does not launder a self-seal, and "the first signer was someone else" is not the clause being

checked. Only a seal by a different legal person, or a qualified time stamp, is out of their reach — and even then the trust service attests only what it was handed (§9.3).
- **(ii) A single seal does not stop a re-forged history.** The operator can doctor a package and have *that* sealed. What defeats it is custody of prior packages by the other party, so the seal multiplies the chain and does not replace it (§9.3).
- **(iii) The legal presumption that attaches to a qualified seal is a presumption about integrity and origin, not a technical guarantee of truth or completeness**, and it must be kept distinct from the residual of §3.3 in any document that cites both (§9.3).

What P3 rests on is in Annex B, and nothing beyond it is claimed here. Item 10 adopts eIDAS (Regulation (EU) No 910/2014) and ETSI EN 319 162-1 as the basis of P3, with the three limits above, and records why the presumption stops where it does: it attaches to the integrity and origin of the sealed data, not to its truth or completeness. Item 11 records eIDAS 2.0 (Regulation (EU) 2024/1183) as introducing the EU Digital Identity Wallet and qualified attestations of attributes about *persons* — and marks the consequence a PROJECTION in the §0.4 sense (MA-4): no change for a package at this version, because the seal, timestamp and preservation instruments P3 uses are the 910/2014 ones (Annex B items 10–11).

What you should not expect

A validator that did not say what it cannot check would be an advertisement. Every validator must admit the following (§10.7 MA-74):

- whether the record is **true**;
- whether the package is **complete** — all that ever existed;
- anything that happened **before the first observation or the first anchor** (§3.5);
- the **producer's binaries**;
- the **honesty of a witness**, including a same-host witness (§3.6);
- **where a witness runs**, or whether it is outside the audited subject's reach: the package can say so in `bound[]` or in its scope, and that statement is the producer's claim, not a checked fact;
- whether a `subject.id` is a **digest** rather than a hostname or an account name: a validator refuses the path shapes it can see and cannot distinguish the rest (§5.2.3);
- **who may read the records**: a payload's own readability classification is a classification by the publisher, and this standard enforces nothing about it — a validator only echoes it.

The same limits seen from the package's side — what a package is not, and the residual every validator prints — are on the first page (index.html#why). In one line: a package proves what it contains was not altered after it left; it cannot prove it is all there

was, and against an operator who rewrote everything before the first anchor it proves nothing.

Where the evidence lives

The repository ships fifty-eight hand-made packages under `vectors/`, each with an `expected.txt` recording the run, the expected exit code, the literal lines it must print, and the `MA-` identifiers the case exercises (MEASURED: 58 directories, 58 `expected.txt` files). None of them is exported from a producer system; they were constructed for the standard (`README.md`).

A standard whose only validator has never rejected anything is a description of its author's output. So the claim that the validator can fail is itself measured: take a copy of `validator/martelage-verify`, delete the one line that compares a record's `prev` to its predecessor's `body_sha` (line 828 of the file as shipped), leave everything else alone — `sh -n` on the mutant is clean — and run the vector built for exactly that check. MEASURED on this tree, both runs trimmed to the refusal block and the verdict:

```
# as shipped
$ sh validator/martelage-verify vectors/02-broken-predecessor/pkg
refusals
  payload: records.jsonl stream 'note': prev at seq 2 does not name seq 1;

verdict      NON-CONFORMANT
exit=1

# mutant: line 821 deleted
$ sh /tmp/noprev vectors/02-broken-predecessor/pkg
refusals
  none

verdict      CONFORMANT
exit=0
```

The mutant still reports `0` on `vectors/01-minimal-conformant` — the mutation removed that check and nothing else (MEASURED). Vector 02 was built for exactly this: its sequence numbers, counts, head and every digest are consistent, so the `prev` comparison is the only check that can catch it. `README.md` lists one such mutant for each check added by review; a check whose removal changes nothing is not load-bearing.

Specification: `spec/MARTELAGE-0.1.md`. Repository and licence: `README.md`, `LICENSE`.

Raising a defect: open one against the *sentence*, not the code — quote the `MA-` identifier and the text, and say what it lets a producer or a verifier claim that the format cannot deliver (see `README.md`, "Raising a defect").